

**NATIONAL UNIVERSITY OF MODERN  
LANGUAGE ISLAMABAD**

**Lab2**



**Submitted to:**

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**Submitted by:**

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**Roll No: ..... SP21379**

**Subject: ..... computer network**

**Department: ..... BSSE-5-Afternoon**

**Date: ..... 10-11-2024**

1. Test the NIC of the PC - Ping 127.0.0.1 called (Loopback address)

```
C:\Users\zulqa>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\zulqa>
```

2. Ping the default gateway.

```
C:\Users\zulqa>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : numlit.edu.pk
    Link-local IPv6 Address . . . . . : fe80::feac:b0a3:d923:4999%9
    IPv4 Address. . . . . : 10.10.90.95
    Subnet Mask . . . . . : 255.255.254.0
    Default Gateway . . . . . : 10.10.90.1

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\zulqa>ping 10.10.90.1

Pinging 10.10.90.1 with 32 bytes of data:
Reply from 10.10.90.1: bytes=32 time=29ms TTL=64
Reply from 10.10.90.1: bytes=32 time=22ms TTL=64
Reply from 10.10.90.1: bytes=32 time=12ms TTL=64
Reply from 10.10.90.1: bytes=32 time=17ms TTL=64

Ping statistics for 10.10.90.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 29ms, Average = 20ms
```

3. Ping your hostname.

```
C:\Users\zulqa> ping zulqarnain

Pinging ZULQARNAIN [fe80::feac:b0a3:d923:499%9] with 32 bytes of data:
Reply from fe80::feac:b0a3:d923:499%9: time<1ms
Reply from fe80::feac:b0a3:d923:499%9: time<1ms
Reply from fe80::feac:b0a3:d923:499%9: time<1ms
Reply from fe80::feac:b0a3:d923:499%9: time<1ms

Ping statistics for fe80::feac:b0a3:d923:499%9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4. Ping the DNS server(s)

```
C:\Users\zulqa>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=38ms TTL=56
Reply from 8.8.8.8: bytes=32 time=44ms TTL=56
Reply from 8.8.8.8: bytes=32 time=37ms TTL=56
Reply from 8.8.8.8: bytes=32 time=39ms TTL=56

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 37ms, Maximum = 44ms, Average = 39ms

C:\Users\zulqa>
```

I get a positive reachability result.

5. How to obtain the following network configuration on your computer.

**Host Name:**

```
C:\Users\zulqa>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : ZULQARNAIN
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Mixed
    IP Routing Enabled . . . . . : No
```

**MAC Address:**

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : E0-2E-0B-EF-8D-A1
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

**IP Address:**

```
Physical Address. . . . . : E0-2E-0B-EF-8D-A0
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::feac:b0a3:d923:499%9(Preferred)
IPv4 Address. . . . . : 10.10.90.95(Preferred)
Subnet Mask . . . . . : 255.255.254.0
Lease Obtained . . . . . : Tuesday, October 15, 2024 9:05:04 PM
```

**Subnet Mask:**

```
Link-local IPv6 Address . . . . . : fe80::feac:b0a3:d923:499%9(Preferred)
IPv4 Address. . . . . : 10.10.90.95(Preferred)
Subnet Mask . . . . . : 255.255.254.0
Lease Obtained. . . . . : Tuesday, October 15, 2024 9:05:04 PM
Lease Expires . . . . . : Tuesday, October 15, 2024 11:15:29 PM
```

**Default Gateway:**

```
Lease Expires . . . . . : Tuesday, October 15, 2024 11:15:29 PM
Default Gateway . . . . . : 10.10.90.1
DHCP Server . . . . . : 192.168.5.98
DHCPv6 IAID . . . . . : 165686795
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-E4-30-77-74-5D-22-44-3B-16
DNS Servers . . . . . : 10.123.30.250
                        10.123.30.252
NetBIOS over Tcpip. . . . . : Enabled
```

**DHCP Servers:**

```
Lease Expires . . . . . : Tuesday, October 15, 2024 11:15:29 PM
Default Gateway . . . . . : 10.10.90.1
DHCP Server . . . . . : 192.168.5.98
DHCPv6 IAID . . . . . : 165686795
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-E4-30-77-74-5D-22-44-3B-16
DNS Servers . . . . . : 10.123.30.250
                        10.123.30.252
```

6. Open a web browser and visit the following website: <http://whatismyipaddress.com/>  
Is the reported IP address consistent with that obtained by ipconfig? If not, why?

My IP Address is:

IPv4: ? **111.68.97.103**

IPv6: ? **Not detected**

```
Connection-specific DNS Suffix  . : numlit.edu.pk
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . : E0-2E-0B-EF-8D-A0
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::feac:b0a3:d923:499%9(Preferred)
IPv4 Address. . . . . : 10.10.90.95(Preferred)
Subnet Mask . . . . . : 255.255.254.0
Lease Obtained. . . . . : Tuesday, October 15, 2024 9:05:04 PM
Lease Expires . . . . . : Tuesday, October 15, 2024 11:15:29 PM
Default Gateway . . . . . : 10.10.90.1
DHCP Server . . . . . : 192.168.5.98
DHCPv6 IAID . . . . . : 165686795
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-E4-30-77-74-5D-22-44-3B-16
DNS Servers . . . . . : 10.123.30.250
                        10.123.30.252
NetBIOS over Tcpip. . . . . : Enabled
```

The reported IP address through above address is different from obtained by ipconfig because, The IP address we see in the ipconfig output is our local (private) IP address assigned by our router.

The IP address shown on the website is our public IP address, which is the address assigned to our network by our Internet Service Provider (ISP). This is the address that other devices on the internet see when we connect to them.

7. Perform ARP and report dynamic addresses.

```
Interface: 10.10.90.95 --- 0x9
Internet Address      Physical Address      Type
10.10.90.1            94-bf-94-5c-3b-c0    dynamic
10.10.90.99           d0-39-57-a6-70-85    dynamic
224.0.0.2             01-00-5e-00-00-02    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

8. Identify the computer names of the dynamic addresses and perform ARP using computer names.